```
import pandas as pd
from IPython.display import display
from sklearn.datasets import load_breast_cancer

breast_cancer_data = load_breast_cancer(as_frame=True)
breast_cancer_df = breast_cancer_data.frame.copy()
breast_cancer_df['diagnosis'] = breast_cancer_df['target'].map({0:
'Malignant', 1: 'Benign'})
print(breast_cancer_df.head())
```

```
   mean radius  mean texture  mean perimeter  mean area  mean
smoothness  \
0        17.99         10.38          122.80     1001.0
0.11840
1        20.57         17.77          132.90     1326.0
0.08474
2        19.69         21.25          130.00     1203.0
0.10960
3        11.42         20.38           77.58      386.1
0.14250
4        20.29         14.34          135.10     1297.0
0.10030

   mean compactness  mean concavity  mean concave points  mean
symmetry  \
0           0.27760          0.3001              0.14710
0.2419
1           0.07864          0.0869              0.07017
0.1812
2           0.15990          0.1974              0.12790
0.2069
3           0.28390          0.2414              0.10520
0.2597
4           0.13280          0.1980              0.10430
0.1809

   mean fractal dimension  ...  worst perimeter  worst area  worst
smoothness  \
0                 0.07871  ...           184.60      2019.0
0.1622
1                 0.05667  ...           158.80      1956.0
0.1238
2                 0.05999  ...           152.50      1709.0
0.1444
3                 0.09744  ...            98.87       567.7
0.2098
4                 0.05883  ...           152.20      1575.0
0.1374

   worst compactness  worst concavity  worst concave points  worst
```

```
symmetry  \
0             0.6656           0.7119                0.2654
0.4601
1             0.1866           0.2416                0.1860
0.2750
2             0.4245           0.4504                0.2430
0.3613
3             0.8663           0.6869                0.2575
0.6638
4             0.2050           0.4000                0.1625
0.2364

   worst fractal dimension  target  diagnosis
0                  0.11890       0  Malignant
1                  0.08902       0  Malignant
2                  0.08758       0  Malignant
3                  0.17300       0  Malignant
4                  0.07678       0  Malignant

[5 rows x 32 columns]
```

```python
import seaborn as sns
import pandas as pd

# in python sns there are some dataset already existing, we can load
them using sns.load_dataset('dataset_name')
# The breast cancer dataset is available from sklearn instead of
seaborn.
df= breast_cancer_df.copy()
print(df.head())
```

```
   mean radius  mean texture  mean perimeter  mean area  mean
smoothness  \
0        17.99         10.38          122.80     1001.0
0.11840
1        20.57         17.77          132.90     1326.0
0.08474
2        19.69         21.25          130.00     1203.0
0.10960
3        11.42         20.38           77.58      386.1
0.14250
4        20.29         14.34          135.10     1297.0
0.10030

   mean compactness  mean concavity  mean concave points  mean
symmetry  \
0           0.27760          0.3001              0.14710
0.2419
1           0.07864          0.0869              0.07017
0.1812
```

```
2           0.15990          0.1974              0.12790
0.2069
3           0.28390          0.2414              0.10520
0.2597
4           0.13280          0.1980              0.10430
0.1809

   mean fractal dimension  ...  worst perimeter  worst area  worst
smoothness  \
0                 0.07871  ...           184.60      2019.0
0.1622
1                 0.05667  ...           158.80      1956.0
0.1238
2                 0.05999  ...           152.50      1709.0
0.1444
3                 0.09744  ...            98.87       567.7
0.2098
4                 0.05883  ...           152.20      1575.0
0.1374

   worst compactness  worst concavity  worst concave points  worst
symmetry  \
0             0.6656           0.7119                0.2654
0.4601
1             0.1866           0.2416                0.1860
0.2750
2             0.4245           0.4504                0.2430
0.3613
3             0.8663           0.6869                0.2575
0.6638
4             0.2050           0.4000                0.1625
0.2364

   worst fractal dimension  target  diagnosis
0                  0.11890       0  Malignant
1                  0.08902       0  Malignant
2                  0.08758       0  Malignant
3                  0.17300       0  Malignant
4                  0.07678       0  Malignant

[5 rows x 32 columns]
```

```
#1 What is the size of the dataset?

print("Dataset shape: ", df.shape)
print("Number of row: ", df.shape[0])
print("Number of column: ", df.shape[1])

print("Dimention of the dataset: ", df.ndim)
```

```
Dataset shape:  (569, 32)
Number of row:  569
Number of column:  32
Dimention of the dataset:  2

#2 How does the dataset look like?
print(df.head())
print(df.tail(10))

   mean radius  mean texture  mean perimeter  mean area  mean
smoothness  \
0        17.99         10.38          122.80     1001.0
0.11840
1        20.57         17.77          132.90     1326.0
0.08474
2        19.69         21.25          130.00     1203.0
0.10960
3        11.42         20.38           77.58      386.1
0.14250
4        20.29         14.34          135.10     1297.0
0.10030

   mean compactness  mean concavity  mean concave points  mean
symmetry  \
0           0.27760          0.3001              0.14710
0.2419
1           0.07864          0.0869              0.07017
0.1812
2           0.15990          0.1974              0.12790
0.2069
3           0.28390          0.2414              0.10520
0.2597
4           0.13280          0.1980              0.10430
0.1809

   mean fractal dimension  ...  worst perimeter  worst area  worst
smoothness  \
0                 0.07871  ...           184.60      2019.0
0.1622
1                 0.05667  ...           158.80      1956.0
0.1238
2                 0.05999  ...           152.50      1709.0
0.1444
3                 0.09744  ...            98.87       567.7
0.2098
4                 0.05883  ...           152.20      1575.0
0.1374

   worst compactness  worst concavity  worst concave points  worst
symmetry  \
```

```
0             0.6656           0.7119                0.2654
0.4601
1             0.1866           0.2416                0.1860
0.2750
2             0.4245           0.4504                0.2430
0.3613
3             0.8663           0.6869                0.2575
0.6638
4             0.2050           0.4000                0.1625
0.2364

   worst fractal dimension  target  diagnosis
0                  0.11890       0  Malignant
1                  0.08902       0  Malignant
2                  0.08758       0  Malignant
3                  0.17300       0  Malignant
4                  0.07678       0  Malignant

[5 rows x 32 columns]
     mean radius  mean texture  mean perimeter  mean area  mean
smoothness  \
559        11.51         23.93           74.52      403.5
0.09261
560        14.05         27.15           91.38      600.4
0.09929
561        11.20         29.37           70.67      386.0
0.07449
562        15.22         30.62          103.40      716.9
0.10480
563        20.92         25.09          143.00     1347.0
0.10990
564        21.56         22.39          142.00     1479.0
0.11100
565        20.13         28.25          131.20     1261.0
0.09780
566        16.60         28.08          108.30      858.1
0.08455
567        20.60         29.33          140.10     1265.0
0.11780
568         7.76         24.54           47.92      181.0
0.05263

     mean compactness  mean concavity  mean concave points  mean
symmetry  \
559           0.10210         0.11120              0.04105
0.1388
560           0.11260         0.04462              0.04304
0.1537
561           0.03558         0.00000              0.00000
0.1060
```

```
562           0.20870          0.25500               0.09429
0.2128
563           0.22360          0.31740               0.14740
0.2149
564           0.11590          0.24390               0.13890
0.1726
565           0.10340          0.14400               0.09791
0.1752
566           0.10230          0.09251               0.05302
0.1590
567           0.27700          0.35140               0.15200
0.2397
568           0.04362          0.00000               0.00000
0.1587

     mean fractal dimension  ...  worst perimeter  worst area  \
559                 0.06570  ...            82.28       474.2
560                 0.06171  ...           100.20       706.7
561                 0.05502  ...            75.19       439.6
562                 0.07152  ...           128.70       915.0
563                 0.06879  ...           179.10      1819.0
564                 0.05623  ...           166.10      2027.0
565                 0.05533  ...           155.00      1731.0
566                 0.05648  ...           126.70      1124.0
567                 0.07016  ...           184.60      1821.0
568                 0.05884  ...            59.16       268.6

     worst smoothness  worst compactness  worst concavity  \
559           0.12980            0.25170           0.3630
560           0.12410            0.22640           0.1326
561           0.09267            0.05494           0.0000
562           0.14170            0.79170           1.1700
563           0.14070            0.41860           0.6599
564           0.14100            0.21130           0.4107
565           0.11660            0.19220           0.3215
566           0.11390            0.30940           0.3403
567           0.16500            0.86810           0.9387
568           0.08996            0.06444           0.0000

     worst concave points  worst symmetry  worst fractal dimension
target  \
559               0.09653          0.2112                  0.08732
1
560               0.10480          0.2250                  0.08321
1
561               0.00000          0.1566                  0.05905
1
562               0.23560          0.4089                  0.14090
0
563               0.25420          0.2929                  0.09873
```

```
0
564                0.22160           0.2060                     0.07115
0
565                0.16280           0.2572                     0.06637
0
566                0.14180           0.2218                     0.07820
0
567                0.26500           0.4087                     0.12400
0
568                0.00000           0.2871                     0.07039
1

     diagnosis
559     Benign
560     Benign
561     Benign
562  Malignant
563  Malignant
564  Malignant
565  Malignant
566  Malignant
567  Malignant
568     Benign

[10 rows x 32 columns]
```

```
#print any random row
df.sample()
df.sample(10)
df.sample(frac=0.1)
```

```
     mean radius  mean texture  mean perimeter  mean area  mean
smoothness  \
455       13.380         30.72           86.34      557.2
0.09245
390       10.260         12.22           65.75      321.6
0.09996
58        13.050         19.31           82.61      527.2
0.08060
456       11.630         29.29           74.87      415.1
0.09357
0         17.990         10.38          122.80     1001.0
0.11840
38        14.990         25.20           95.54      698.8
0.09387
149       13.740         17.91           88.12      585.0
0.07944
428       11.130         16.62           70.47      381.1
0.08151
310       11.700         19.11           74.33      418.7
```

```
0.08814
418       12.700          12.17           80.88       495.0
0.08785
415       11.890          21.17           76.39       433.8
0.09773
194       14.860          23.21          100.40       671.4
0.10440
95        20.260          23.03          132.40      1264.0
0.09078
528       13.940          13.17           90.31       594.2
0.12480
108       22.270          19.67          152.80      1509.0
0.13260
91        15.370          22.76          100.20       728.2
0.09200
393       21.610          22.28          144.40      1407.0
0.11670
379       11.080          18.83           73.30       361.6
0.12160
357       13.870          16.21           88.52       593.7
0.08743
435       13.980          19.62           91.12       599.5
0.10600
369       22.010          21.90          147.20      1482.0
0.10630
44        13.170          21.81           85.42       531.5
0.09714
427       10.800          21.98           68.79       359.9
0.08801
322       12.860          13.32           82.82       504.8
0.11340
43        13.280          20.28           87.32       545.2
0.10410
445       11.990          24.89           77.61       441.3
0.10300
289       11.370          18.89           72.17       396.0
0.08713
129       19.790          25.12          130.40      1192.0
0.10150
190       14.220          23.12           94.37       609.9
0.10750
90        14.620          24.02           94.57       662.7
0.08974
495       14.870          20.21           96.12       680.9
0.09587
470        9.667          18.49           61.49       289.1
0.08946
350       11.660          17.07           73.70       421.0
0.07561
```

```
177        16.460          20.11           109.30       832.9
0.09831
76         13.530          10.94            87.91       559.2
0.12910
2          19.690          21.25           130.00      1203.0
0.10960
408        17.990          20.66           117.80       991.7
0.10360
336        12.990          14.23            84.08       514.3
0.09462
235        14.030          21.25            89.79       603.4
0.09070
382        12.050          22.72            78.75       447.8
0.06935
545        13.620          23.23            87.19       573.2
0.09246
439        14.020          15.66            89.59       606.5
0.07966
467         9.668          18.10            61.06       286.3
0.08311
316        12.180          14.08            77.25       461.4
0.07734
178        13.010          22.22            82.01       526.4
0.06251
164        23.270          22.04           152.10      1686.0
0.08439
356        13.050          18.59            85.09       512.0
0.10820
49         13.490          22.30            86.91       561.0
0.08752
131        15.460          19.48           101.70       748.9
0.10920
568         7.760          24.54            47.92       181.0
0.05263
217        10.200          17.48            65.05       321.2
0.08054
334        12.300          19.02            77.88       464.4
0.08313
416         9.405          21.70            59.60       271.2
0.10440
264        17.190          22.07           111.60       928.3
0.09726
402        12.960          18.29            84.18       525.2
0.07351
292        12.950          16.02            83.14       513.7
0.10050
280        19.160          26.60           126.20      1138.0
0.10200
```

```
     mean compactness  mean concavity  mean concave points  mean
symmetry  \
455           0.07426        0.028190             0.032640
0.1375
390           0.07542        0.019230             0.019680
0.1800
58            0.03789        0.000692             0.004167
0.1819
456           0.08574        0.071600             0.020170
0.1799
0             0.27760        0.300100             0.147100
0.2419
38            0.05131        0.023980             0.028990
0.1565
149           0.06376        0.028810             0.013290
0.1473
428           0.03834        0.013690             0.013700
0.1511
310           0.05253        0.015830             0.011480
0.1936
418           0.05794        0.023600             0.024020
0.1583
415           0.08120        0.025550             0.021790
0.2019
194           0.19800        0.169700             0.088780
0.1737
95            0.13130        0.146500             0.086830
0.2095
528           0.09755        0.101000             0.066150
0.1976
108           0.27680        0.426400             0.182300
0.2556
91            0.10360        0.112200             0.074830
0.1717
393           0.20870        0.281000             0.156200
0.2162
379           0.21540        0.168900             0.063670
0.2196
357           0.05492        0.015020             0.020880
0.1424
435           0.11330        0.112600             0.064630
0.1669
369           0.19540        0.244800             0.150100
0.1824
44            0.10470        0.082590             0.052520
0.1746
427           0.05743        0.036140             0.014040
0.2016
322           0.08834        0.038000             0.034000
```

```
0.1543
43              0.14360         0.098470                0.061580
0.1974
445             0.09218         0.054410                0.042740
0.1820
289             0.05008         0.023990                0.021730
0.2013
129             0.15890         0.254500                0.114900
0.2202
190             0.24130         0.198100                0.066180
0.2384
90              0.08606         0.031020                0.029570
0.1685
495             0.08345         0.068240                0.049510
0.1487
470             0.06258         0.029480                0.015140
0.2238
350             0.03630         0.008306                0.011620
0.1671
177             0.15560         0.179300                0.088660
0.1794
76              0.10470         0.068770                0.065560
0.2403
2               0.15990         0.197400                0.127900
0.2069
408             0.13040         0.120100                0.088240
0.1992
336             0.09965         0.037380                0.020980
0.1652
235             0.06945         0.014620                0.018960
0.1517
382             0.10730         0.079430                0.029780
0.1203
545             0.06747         0.029740                0.024430
0.1664
439             0.05581         0.020870                0.026520
0.1589
467             0.05428         0.014790                0.005769
0.1680
316             0.03212         0.011230                0.005051
0.1673
178             0.01938         0.001595                0.001852
0.1395
164             0.11450         0.132400                0.097020
0.1801
356             0.13040         0.096030                0.056030
0.2035
49              0.07698         0.047510                0.033840
0.1809
```

```
131           0.12230         0.146600              0.080870
0.1931
568           0.04362         0.000000              0.000000
0.1587
217           0.05907         0.057740              0.010710
0.1964
334           0.04202         0.007756              0.008535
0.1539
416           0.06159         0.020470              0.012570
0.2025
264           0.08995         0.090610              0.065270
0.1867
402           0.07899         0.040570              0.018830
0.1874
292           0.07943         0.061550              0.033700
0.1730
280           0.14530         0.192100              0.096640
0.1902

     mean fractal dimension  ...  worst perimeter  worst area  \
455                 0.06016  ...            96.69       705.6
390                 0.06569  ...            73.23       394.5
58                  0.05501  ...            90.24       624.1
456                 0.06166  ...            86.04       527.8
0                   0.07871  ...           184.60      2019.0
38                  0.05504  ...            95.54       698.8
149                 0.05580  ...            97.19       725.9
428                 0.06148  ...            74.35       421.1
310                 0.06128  ...            80.92       483.1
418                 0.06275  ...            88.12       566.9
415                 0.06290  ...            85.09       522.9
194                 0.06672  ...           118.60       784.7
95                  0.05649  ...           156.10      1750.0
528                 0.06457  ...            94.52       653.3
108                 0.07039  ...           206.80      2360.0
91                  0.06097  ...           107.50       830.9
393                 0.06606  ...           172.00      2081.0
379                 0.07950  ...            91.76       508.1
357                 0.05883  ...            96.74       694.4
435                 0.06544  ...           113.90       869.3
369                 0.06140  ...           195.00      2227.0
44                  0.06177  ...           105.50       740.7
427                 0.05977  ...            83.69       489.5
322                 0.06476  ...            92.80       599.5
43                  0.06782  ...           113.10       907.2
445                 0.06850  ...            84.48       513.9
289                 0.05955  ...            79.29       459.3
129                 0.06113  ...           148.70      1589.0
190                 0.07542  ...           106.40       762.4
```

```
90                  0.05866  ...           102.90       803.7
495                 0.05748  ...           103.90       783.6
470                 0.06413  ...            70.88       385.2
350                 0.05731  ...            83.61       542.5
177                 0.06323  ...           123.50       981.2
76                  0.06641  ...            91.36       605.5
2                   0.05999  ...           152.50      1709.0
408                 0.06069  ...           138.10      1349.0
336                 0.07238  ...            87.38       576.0
235                 0.05835  ...            98.27       715.5
382                 0.06659  ...            87.36       488.4
545                 0.05801  ...            97.58       729.8
439                 0.05586  ...            96.53       688.9
467                 0.06412  ...            71.11       380.2
316                 0.05649  ...            81.60       513.1
178                 0.05234  ...            88.18       608.8
164                 0.05553  ...           184.20      2403.0
356                 0.06501  ...            94.22       591.2
49                  0.05718  ...            99.00       698.8
131                 0.05796  ...           124.90      1156.0
568                 0.05884  ...            59.16       268.6
217                 0.06315  ...            75.40       403.7
334                 0.05945  ...            84.53       544.3
416                 0.06601  ...            68.73       359.4
264                 0.05580  ...           140.50      1436.0
402                 0.05899  ...            96.31       621.9
292                 0.06470  ...            88.81       585.4
280                 0.06220  ...           159.80      1724.0

     worst smoothness  worst compactness  worst concavity  \
455           0.11720            0.14210         0.070030
390           0.13430            0.16500         0.086150
58            0.10210            0.06191         0.001845
456           0.14060            0.20310         0.292300
0             0.16220            0.66560         0.711900
38            0.09387            0.05131         0.023980
149           0.09711            0.18240         0.156400
428           0.10300            0.06219         0.045800
310           0.12230            0.10870         0.079150
418           0.13140            0.16070         0.093850
415           0.14260            0.21870         0.116400
194           0.13160            0.46480         0.458900
95            0.11900            0.35390         0.409800
528           0.13940            0.13640         0.155900
108           0.17010            0.69970         0.960800
91            0.12570            0.19970         0.284600
393           0.15020            0.57170         0.705300
379           0.21840            0.93790         0.840200
357           0.11530            0.10080         0.052850
```

```
435            0.16130            0.35680          0.406900
369            0.12940            0.38850          0.475600
44             0.15030            0.39040          0.372800
427            0.13030            0.16960          0.192700
322            0.15470            0.22310          0.179100
43             0.15300            0.37240          0.366400
445            0.13110            0.18220          0.160900
289            0.11180            0.09708          0.075290
129            0.12750            0.38610          0.567300
190            0.15330            0.93270          0.848800
90             0.11150            0.17660          0.091890
495            0.12160            0.13880          0.170000
470            0.12340            0.15420          0.127700
350            0.09958            0.06476          0.030460
177            0.14150            0.46670          0.586200
76             0.14510            0.13790          0.085390
2              0.14440            0.42450          0.450400
408            0.14820            0.37350          0.330100
336            0.11420            0.19750          0.145000
235            0.12870            0.15130          0.062310
382            0.08799            0.32140          0.291200
545            0.12160            0.15170          0.104900
439            0.10340            0.10170          0.062600
467            0.13880            0.12550          0.064090
316            0.10010            0.05332          0.041160
178            0.08125            0.03432          0.007977
164            0.12280            0.35830          0.394800
356            0.13430            0.26580          0.257300
49             0.11620            0.17110          0.228200
131            0.15460            0.23940          0.379100
568            0.08996            0.06444          0.000000
217            0.09527            0.13970          0.192500
334            0.12220            0.09052          0.036190
416            0.15260            0.11930          0.061410
264            0.15580            0.25670          0.388900
402            0.09329            0.23180          0.160400
292            0.14830            0.20680          0.224100
280            0.17820            0.38410          0.575400

     worst concave points  worst symmetry  worst fractal dimension
target  \
455               0.077630          0.2196                  0.07675
1
390               0.066960          0.2937                  0.07722
1
58                0.011110          0.2439                  0.06289
1
456               0.068350          0.2884                  0.07220
1
```

0	0.265400	0.4601	0.11890
0			
38	0.028990	0.1565	0.05504
0			
149	0.060190	0.2350	0.07014
1			
428	0.040440	0.2383	0.07083
1			
310	0.057410	0.3487	0.06958
1			
418	0.082240	0.2775	0.09464
1			
415	0.082630	0.3075	0.07351
1			
194	0.172700	0.3000	0.08701
0			
95	0.157300	0.3689	0.08368
0			
528	0.101500	0.2160	0.07253
1			
108	0.291000	0.4055	0.09789
0			
91	0.147600	0.2556	0.06828
0			
393	0.242200	0.3828	0.10070
0			
379	0.252400	0.4154	0.14030
0			
357	0.055560	0.2362	0.07113
1			
435	0.182700	0.3179	0.10550
0			
369	0.243200	0.2741	0.08574
0			
44	0.160700	0.3693	0.09618
0			
427	0.074850	0.2965	0.07662
1			
322	0.115500	0.2382	0.08553
1			
43	0.149200	0.3739	0.10270
0			
445	0.120200	0.2599	0.08251
1			
289	0.062030	0.3267	0.06994
1			
129	0.173200	0.3305	0.08465
0			
190	0.177200	0.5166	0.14460

```
0
90                  0.069460            0.2522                  0.07246
1
495                 0.101700            0.2369                  0.06599
1
470                 0.065600            0.3174                  0.08524
1
350                 0.042620            0.2731                  0.06825
1
177                 0.203500            0.3054                  0.09519
0
76                  0.074070            0.2710                  0.07191
1
2                   0.243000            0.3613                  0.08758
0
408                 0.197400            0.3060                  0.08503
0
336                 0.058500            0.2432                  0.10090
1
235                 0.079630            0.2226                  0.07617
1
382                 0.109200            0.2191                  0.09349
1
545                 0.071740            0.2642                  0.06953
1
439                 0.082160            0.2136                  0.06710
1
467                 0.025000            0.3057                  0.07875
1
316                 0.018520            0.2293                  0.06037
1
178                 0.009259            0.2295                  0.05843
1
164                 0.234600            0.3589                  0.09187
0
356                 0.125800            0.3113                  0.08317
1
49                  0.128200            0.2871                  0.06917
1
131                 0.151400            0.2837                  0.08019
0
568                 0.000000            0.2871                  0.07039
1
217                 0.035710            0.2868                  0.07809
1
334                 0.039830            0.2554                  0.07207
1
416                 0.037700            0.2872                  0.08304
1
```

```
264              0.198400          0.3216                   0.07570
0
402              0.066080          0.3207                   0.07247
1
292              0.105600          0.3380                   0.09584
1
280              0.187200          0.3258                   0.09720
0

     diagnosis
455     Benign
390     Benign
58      Benign
456     Benign
0    Malignant
38   Malignant
149     Benign
428     Benign
310     Benign
418     Benign
415     Benign
194  Malignant
95   Malignant
528     Benign
108  Malignant
91   Malignant
393  Malignant
379  Malignant
357     Benign
435  Malignant
369  Malignant
44   Malignant
427     Benign
322     Benign
43   Malignant
445     Benign
289     Benign
129  Malignant
190  Malignant
90      Benign
495     Benign
470     Benign
350     Benign
177  Malignant
76      Benign
2    Malignant
408  Malignant
336     Benign
235     Benign
382     Benign
```

```
545     Benign
439     Benign
467     Benign
316     Benign
178     Benign
164  Malignant
356     Benign
49      Benign
131  Malignant
568     Benign
217     Benign
334     Benign
416     Benign
264  Malignant
402     Benign
292     Benign
280  Malignant

[57 rows x 32 columns]
```

```
df.tail(7)
```

```
     mean radius  mean texture  mean perimeter  mean area  mean
smoothness  \
562        15.22         30.62          103.40      716.9
0.10480
563        20.92         25.09          143.00     1347.0
0.10990
564        21.56         22.39          142.00     1479.0
0.11100
565        20.13         28.25          131.20     1261.0
0.09780
566        16.60         28.08          108.30      858.1
0.08455
567        20.60         29.33          140.10     1265.0
0.11780
568         7.76         24.54           47.92      181.0
0.05263

     mean compactness  mean concavity  mean concave points  mean
symmetry  \
562           0.20870         0.25500              0.09429
0.2128
563           0.22360         0.31740              0.14740
0.2149
564           0.11590         0.24390              0.13890
0.1726
565           0.10340         0.14400              0.09791
0.1752
566           0.10230         0.09251              0.05302
```

```
0.1590
567           0.27700         0.35140              0.15200
0.2397
568           0.04362         0.00000              0.00000
0.1587

     mean fractal dimension  ...  worst perimeter  worst area  \
562                 0.07152  ...           128.70       915.0
563                 0.06879  ...           179.10      1819.0
564                 0.05623  ...           166.10      2027.0
565                 0.05533  ...           155.00      1731.0
566                 0.05648  ...           126.70      1124.0
567                 0.07016  ...           184.60      1821.0
568                 0.05884  ...            59.16       268.6

     worst smoothness  worst compactness  worst concavity  \
562           0.14170            0.79170           1.1700
563           0.14070            0.41860           0.6599
564           0.14100            0.21130           0.4107
565           0.11660            0.19220           0.3215
566           0.11390            0.30940           0.3403
567           0.16500            0.86810           0.9387
568           0.08996            0.06444           0.0000

     worst concave points  worst symmetry  worst fractal dimension
target  \
562                0.2356          0.4089                  0.14090
0
563                0.2542          0.2929                  0.09873
0
564                0.2216          0.2060                  0.07115
0
565                0.1628          0.2572                  0.06637
0
566                0.1418          0.2218                  0.07820
0
567                0.2650          0.4087                  0.12400
0
568                0.0000          0.2871                  0.07039
1

     diagnosis
562  Malignant
563  Malignant
564  Malignant
565  Malignant
566  Malignant
567  Malignant
568     Benign
```

```
[7 rows x 32 columns]
```

```
#3 What is the data type of each column?
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 569 entries, 0 to 568
Data columns (total 32 columns):
 #   Column                   Non-Null Count  Dtype  
---  ------                   --------------  -----  
 0   mean radius              569 non-null    float64
 1   mean texture             569 non-null    float64
 2   mean perimeter           569 non-null    float64
 3   mean area                569 non-null    float64
 4   mean smoothness          569 non-null    float64
 5   mean compactness         569 non-null    float64
 6   mean concavity           569 non-null    float64
 7   mean concave points      569 non-null    float64
 8   mean symmetry            569 non-null    float64
 9   mean fractal dimension   569 non-null    float64
 10  radius error             569 non-null    float64
 11  texture error            569 non-null    float64
 12  perimeter error          569 non-null    float64
 13  area error               569 non-null    float64
 14  smoothness error         569 non-null    float64
 15  compactness error        569 non-null    float64
 16  concavity error          569 non-null    float64
 17  concave points error     569 non-null    float64
 18  symmetry error           569 non-null    float64
 19  fractal dimension error  569 non-null    float64
 20  worst radius             569 non-null    float64
 21  worst texture            569 non-null    float64
 22  worst perimeter          569 non-null    float64
 23  worst area               569 non-null    float64
 24  worst smoothness         569 non-null    float64
 25  worst compactness        569 non-null    float64
 26  worst concavity          569 non-null    float64
 27  worst concave points     569 non-null    float64
 28  worst symmetry           569 non-null    float64
 29  worst fractal dimension  569 non-null    float64
 30  target                   569 non-null    int64  
 31  diagnosis                569 non-null    object 
dtypes: float64(30), int64(1), object(1)
memory usage: 142.4+ KB
```

```
display(df.columns)
```

```
Index(['mean radius', 'mean texture', 'mean perimeter', 'mean area',
       'mean smoothness', 'mean compactness', 'mean concavity',
       'mean concave points', 'mean symmetry', 'mean fractal
dimension',
       'radius error', 'texture error', 'perimeter error', 'area
error',
       'smoothness error', 'compactness error', 'concavity error',
       'concave points error', 'symmetry error', 'fractal dimension
error',
       'worst radius', 'worst texture', 'worst perimeter', 'worst
area',
       'worst smoothness', 'worst compactness', 'worst concavity',
       'worst concave points', 'worst symmetry', 'worst fractal
dimension',
       'target', 'diagnosis'],
      dtype='object')
```

```
#4 Are there any missing values in the dataset?

df.isnull().sum()
```

```
mean radius                 0
mean texture                0
mean perimeter              0
mean area                   0
mean smoothness             0
mean compactness            0
mean concavity              0
mean concave points         0
mean symmetry               0
mean fractal dimension      0
radius error                0
texture error               0
perimeter error             0
area error                  0
smoothness error            0
compactness error           0
concavity error             0
concave points error        0
symmetry error              0
fractal dimension error     0
worst radius                0
worst texture               0
worst perimeter             0
worst area                  0
worst smoothness            0
worst compactness           0
worst concavity             0
worst concave points        0
worst symmetry              0
```

```
worst fractal dimension    0
target                     0
diagnosis                  0
dtype: int64
```

```
#5 How does the dataset look mathematically? (mean, median, mode,
standard deviation, etc.)

display(df.describe())
```

```
       mean radius  mean texture  mean perimeter    mean area  \
count   569.000000    569.000000      569.000000   569.000000
mean     14.127292     19.289649       91.969033   654.889104
std       3.524049      4.301036       24.298981   351.914129
min       6.981000      9.710000       43.790000   143.500000
25%      11.700000     16.170000       75.170000   420.300000
50%      13.370000     18.840000       86.240000   551.100000
75%      15.780000     21.800000      104.100000   782.700000
max      28.110000     39.280000      188.500000  2501.000000

       mean smoothness  mean compactness  mean concavity  mean concave
points  \
count       569.000000        569.000000      569.000000
569.000000
mean          0.096360          0.104341        0.088799
0.048919
std           0.014064          0.052813        0.079720
0.038803
min           0.052630          0.019380        0.000000
0.000000
25%           0.086370          0.064920        0.029560
0.020310
50%           0.095870          0.092630        0.061540
0.033500
75%           0.105300          0.130400        0.130700
0.074000
max           0.163400          0.345400        0.426800
0.201200

       mean symmetry  mean fractal dimension  ...  worst texture  \
count     569.000000              569.000000  ...     569.000000
mean        0.181162                0.062798  ...      25.677223
std         0.027414                0.007060  ...       6.146258
min         0.106000                0.049960  ...      12.020000
25%         0.161900                0.057700  ...      21.080000
50%         0.179200                0.061540  ...      25.410000
75%         0.195700                0.066120  ...      29.720000
max         0.304000                0.097440  ...      49.540000

       worst perimeter   worst area  worst smoothness  worst
```

```
compactness  \
count       569.000000   569.000000         569.000000
569.000000
mean        107.261213   880.583128           0.132369
0.254265
std          33.602542   569.356993           0.022832
0.157336
min          50.410000   185.200000           0.071170
0.027290
25%          84.110000   515.300000           0.116600
0.147200
50%          97.660000   686.500000           0.131300
0.211900
75%         125.400000  1084.000000           0.146000
0.339100
max         251.200000  4254.000000           0.222600
1.058000

       worst concavity  worst concave points  worst symmetry  \
count       569.000000            569.000000      569.000000
mean          0.272188              0.114606        0.290076
std           0.208624              0.065732        0.061867
min           0.000000              0.000000        0.156500
25%           0.114500              0.064930        0.250400
50%           0.226700              0.099930        0.282200
75%           0.382900              0.161400        0.317900
max           1.252000              0.291000        0.663800

       worst fractal dimension      target
count                569.000000  569.000000
mean                   0.083946    0.627417
std                    0.018061    0.483918
min                    0.055040    0.000000
25%                    0.071460    0.000000
50%                    0.080040    1.000000
75%                    0.092080    1.000000
max                    0.207500    1.000000

[8 rows x 31 columns]
```

```
print(df.describe().loc['mean', 'mean radius'])
print(df['mean radius'].mean())
```

```
14.127291739894552
14.127291739894552
```

```
#6 Are there any duplicate values?
print(df.duplicated().sum())
```

```
0
```

```
#remove all duplicated values

df=df.drop_duplicates()
df.reset_index(drop=True, inplace=True)
df.duplicated().sum()
```

```
np.int64(0)
```

```
#7 How is the correlation among columns?

#df.select_dtypes(include=['float64', 'int64']).corr()
#non column correlation
df.select_dtypes(include=['float64', 'int64']).corrwith(df['mean
radius'])
```

```
mean radius                1.000000
mean texture               0.323782
mean perimeter             0.997855
mean area                  0.987357
mean smoothness            0.170581
mean compactness           0.506124
mean concavity             0.676764
mean concave points        0.822529
mean symmetry              0.147741
mean fractal dimension    -0.311631
radius error               0.679090
texture error             -0.097317
perimeter error            0.674172
area error                 0.735864
smoothness error          -0.222600
compactness error          0.206000
concavity error            0.194204
concave points error       0.376169
symmetry error            -0.104321
fractal dimension error   -0.042641
worst radius               0.969539
worst texture              0.297008
worst perimeter            0.965137
worst area                 0.941082
worst smoothness           0.119616
worst compactness          0.413463
worst concavity            0.526911
worst concave points       0.744214
worst symmetry             0.163953
worst fractal dimension    0.007066
target                    -0.730029
dtype: float64
```

```
import numpy as np
df.select_dtypes(include=[np.number]).corr()['target']
```

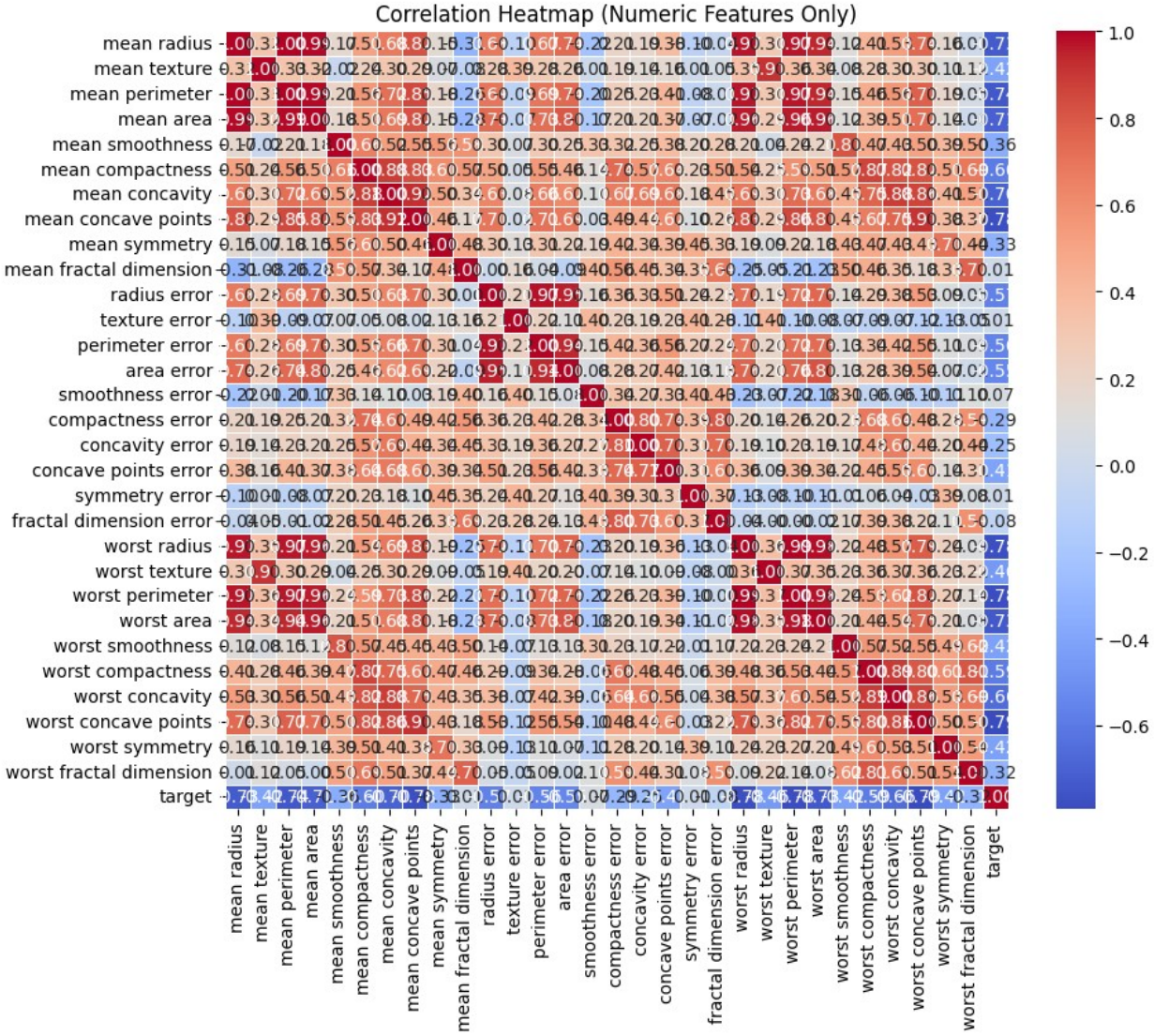
```
mean radius                -0.730029
mean texture               -0.415185
mean perimeter             -0.742636
mean area                  -0.708984
mean smoothness            -0.358560
mean compactness           -0.596534
mean concavity             -0.696360
mean concave points        -0.776614
mean symmetry              -0.330499
mean fractal dimension      0.012838
radius error               -0.567134
texture error               0.008303
perimeter error            -0.556141
area error                 -0.548236
smoothness error            0.067016
compactness error          -0.292999
concavity error            -0.253730
concave points error       -0.408042
symmetry error              0.006522
fractal dimension error    -0.077972
worst radius               -0.776454
worst texture              -0.456903
worst perimeter            -0.782914
worst area                 -0.733825
worst smoothness           -0.421465
worst compactness          -0.590998
worst concavity            -0.659610
worst concave points       -0.793566
worst symmetry             -0.416294
worst fractal dimension    -0.323872
target                      1.000000
Name: target, dtype: float64
```

```python
import matplotlib.pyplot as plt

numeric_df= df.select_dtypes(include='number')
correlations= numeric_df.corr(method='pearson')

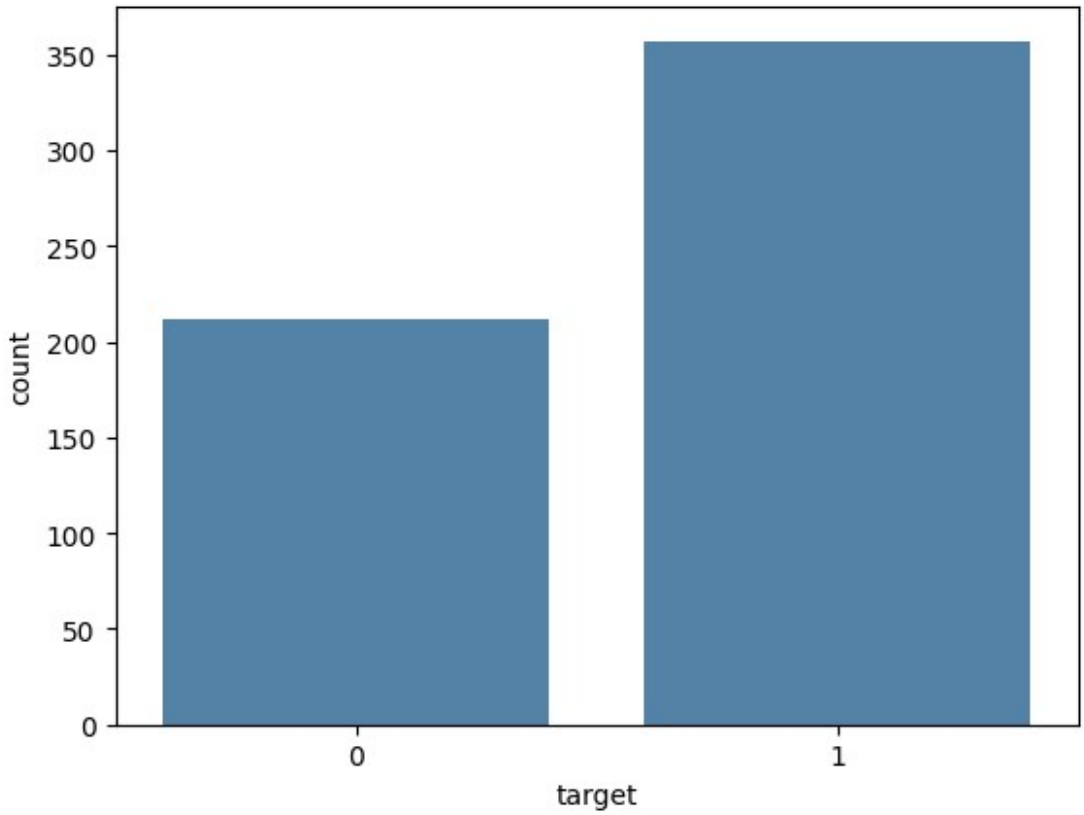
plt.figure(figsize=(10,8))
sns.heatmap(correlations, annot=True, fmt='.2f', cmap='coolwarm',
linewidths=0.5)

plt.title('Correlation Heatmap (Numeric Features Only)')
plt.show()
```




```
sns.countplot(df, x='target', color='steelblue')
```

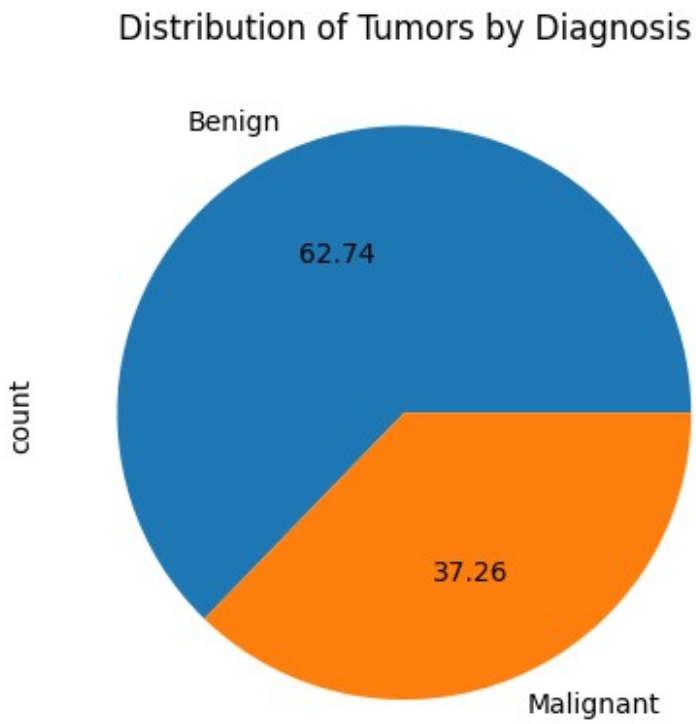
```
<Axes: xlabel='target', ylabel='count'>
```



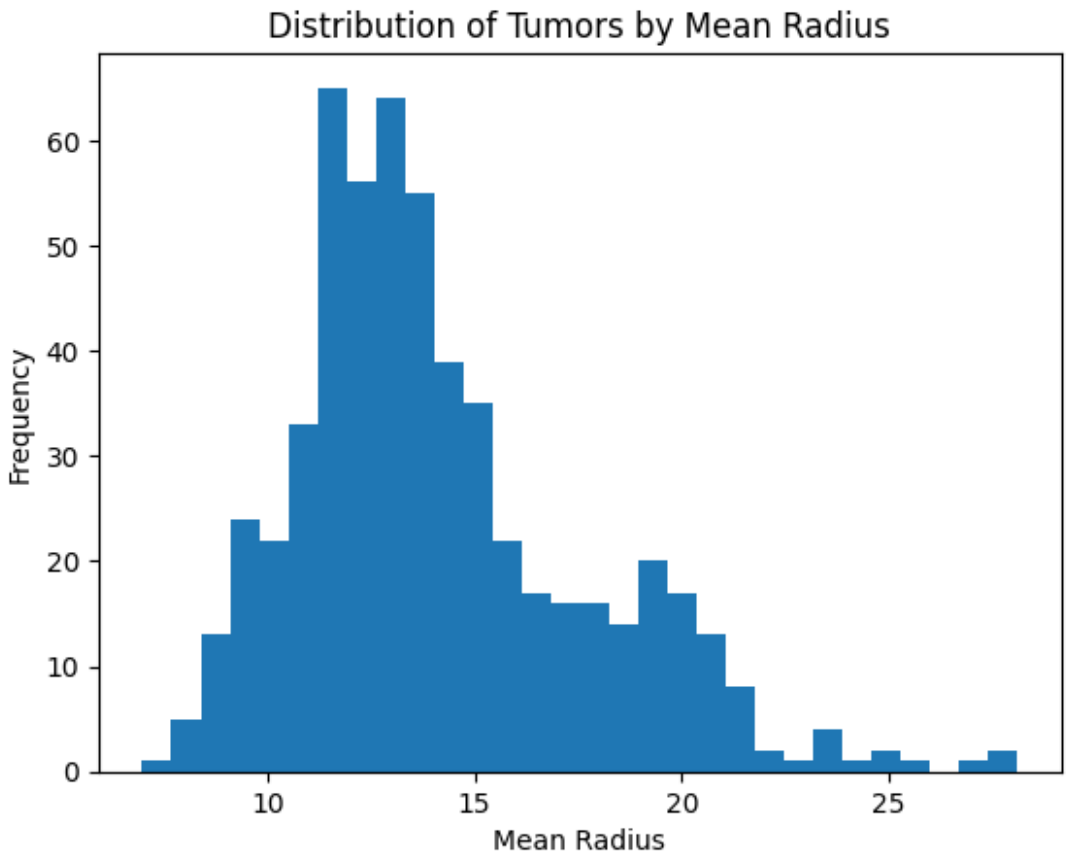

```
df['target'].value_counts()
```

```
target
1    357
0    212
Name: count, dtype: int64
```

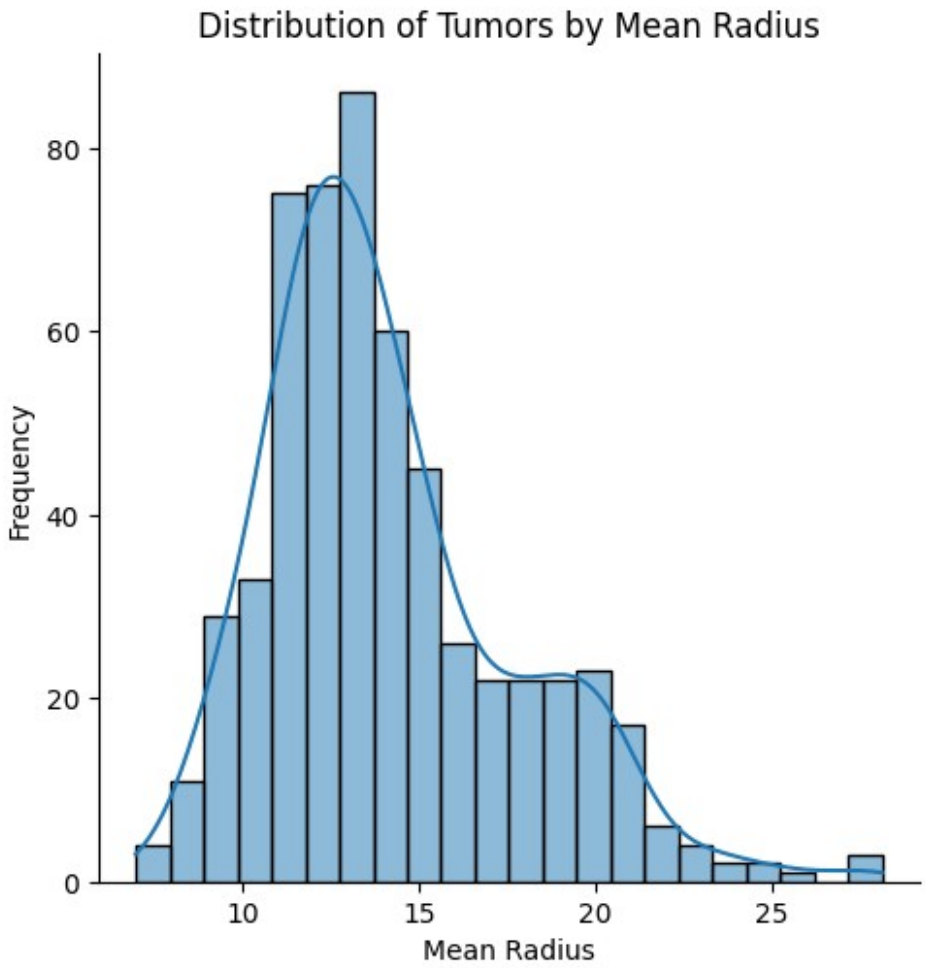
```
df['diagnosis'].value_counts().plot(kind='pie', autopct='%.2f')
plt.title('Distribution of Tumors by Diagnosis')
plt.show()
```




```
plt.hist(df['mean radius'], bins=30)
plt.xlabel('Mean Radius')
plt.ylabel('Frequency')
plt.title('Distribution of Tumors by Mean Radius')
plt.show()
```

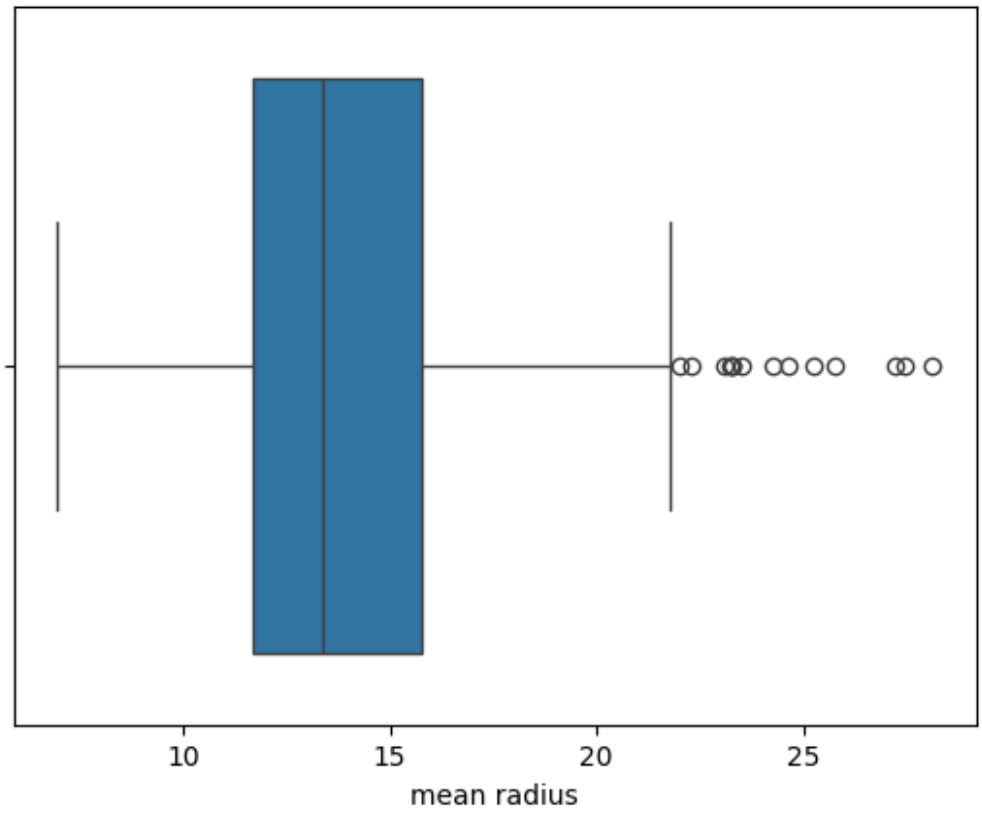



```
sns.displot(df['mean radius'], kde=True)
plt.xlabel('Mean Radius')
plt.ylabel('Frequency')
plt.title('Distribution of Tumors by Mean Radius')
plt.show()
```




```
sns.boxplot(x='mean radius', data=df)
```

```
<Axes: xlabel='mean radius'>
```

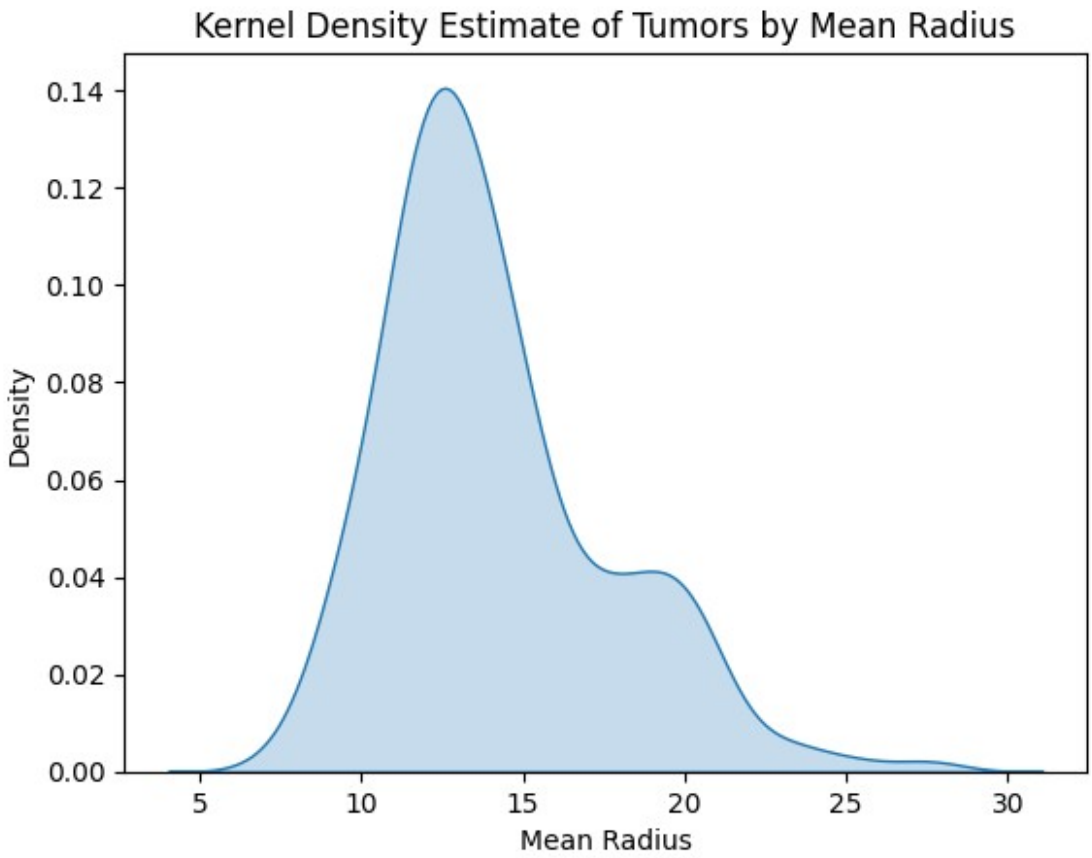



```
sns.kdeplot(df['mean radius'], shade=True)
plt.xlabel('Mean Radius')
plt.ylabel('Density')
plt.title('Kernel Density Estimate of Tumors by Mean Radius')
plt.show()
```

```
C:\Users\mdfah\AppData\Local\Temp\ipykernel_21640\1042504216.py:1:
FutureWarning:

`shade` is now deprecated in favor of `fill`; setting `fill=True`.
This will become an error in seaborn v0.14.0; please update your code.

  sns.kdeplot(df['mean radius'], shade=True)
```




```
df['mean radius'].skew()
```

```
np.float64(0.9423795716730992)
```